DRUGS · THYROID AND PARATHYROID

Cinacalcet

Cinacalcet is a calcimimetic: it increases the sensitivity of the parathyroid calcium-sensing receptor so that the glands secrete less PTH at a given calcium. It lowers calcium and PTH in primary hyperparathyroidism when surgery is not possible or has failed, in parathyroid carcinoma, and in secondary hyperparathyroidism in patients on dialysis. It does not improve bone density in primary disease, so surgery remains the definitive treatment.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

CINACALCET

	SUMMARY
Class	Calcimimetic (allosteric modulator of the calcium-sensing receptor)
ENT uses	Primary hyperparathyroidism when parathyroidectomy is contraindicated, refused or has failed; parathyroid carcinoma with hypercalcaemia; secondary hyperparathyroidism in end-stage kidney disease (alongside or instead of parathyroidectomy); before surgery in severe hypercalcaemia
Dose	Primary hyperparathyroidism and carcinoma: 30 mg twice daily, titrated up to 90 mg three or four times daily; secondary hyperparathyroidism: 30 mg once daily, titrated to a maximum of 180 mg daily
Monitoring	Calcium within a week of starting or changing the dose, then regularly; PTH
Etelcalcetide	An intravenous calcimimetic given at dialysis

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Primary hyperparathyroidism

Surgery is recommended for patients who meet guideline criteria (symptomatic, calcium more than 0.25 mmol/l above the upper limit, osteoporosis or fragility fracture, renal stones or impaired renal function, age under 50) and is curative in about 95% with an experienced surgeon. Cinacalcet controls calcium but does not cure the disease or reverse bone loss.

2.2 Secondary hyperparathyroidism

In dialysis patients, cinacalcet with vitamin D and phosphate binders controls PTH; parathyroidectomy (subtotal or total with autotransplantation) is used when medical treatment fails or causes intolerable effects.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Nausea and vomiting (common), **hypocalcaemia** (paraesthesia, cramps, seizures, prolonged QT), dizziness, rash; gastrointestinal bleeding reported.

3.2 Interactions

A strong **CYP2D6 inhibitor** (raises levels of flecainide, tricyclics, metoprolol and others); levels raised by strong CYP3A4 inhibitors.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism (calcium-sensing receptor).
- Why surgery remains first choice in primary hyperparathyroidism.

4.2 Common pitfalls

COMMON PITFALLS

- Using cinacalcet in place of surgery in a fit patient with osteoporosis.

Written for this app from BNF and BNF for Children; NICE and specialist society guidance; Scott-Brown; Cummings. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.